

Mobile Developer Interview Assignment

Project: Classic Minesweeper Game

Overview

You are required to develop a classic Minesweeper mobile game using the latest versions of React Native and Expo frameworks. The application should run locally on the device without requiring any backend integration.

Core Requirements

- The app must contain two screens: Home Screen and Game Screen.
- Game Screen must display the Minesweeper board and provide two modes: Dig and Flag.
- When a user taps a square, the currently selected mode must be applied.
- If a square is flagged, it must not be possible to dig it until unflagged.
- The game must implement classic Minesweeper logic with mines randomly distributed.
- When a square is revealed, it must display a number (1–8) indicating how many mines surround it in all eight directions.
- If a revealed square has 0 adjacent mines, a chain reaction must automatically reveal neighboring squares until numbered cells are reached.
- The player wins when all safe squares are revealed.
- The player loses immediately upon revealing a mine.

Game Mechanics Details

- Numbers range from 1 to 8 and represent the exact number of mines touching the square (including diagonals).
- Blank squares represent 0 adjacent mines.
- Chain reaction reveal must continue recursively or iteratively until numbered cells are reached.
- Flagging is used to mark suspected mines and is optional for winning.

Bonus Features (Optional but Appreciated)

- Light and Dark Theme support.
- Sound effects for actions (dig, flag, win, lose).
- Settings Screen to enable/disable sound effects.
- Clean animations and smooth user experience.

Technical Expectations

- Use the latest stable version of React Native and Expo.
- Apply clean architecture and maintain separation of concerns.
- Follow best practices for state management and component design.
- Ensure code readability and proper documentation.
- Handle edge cases such as rapid taps and boundary checks properly.

Submission Requirements

- Submit the project as a private Git repository.
- Invite GitHub user: omutas (<https://github.com/omutas>) for review.
- Include a clear README file explaining setup and run instructions.
- Provide an overview of the code structure.
- Document assumptions made and limitations of the implementation.
- Maintain meaningful Git commit history.
- Ensure the project runs without errors.

Good to Have (Not Mandatory)

Provide a short description of your implementation approach, including challenges faced and how you solved them.